

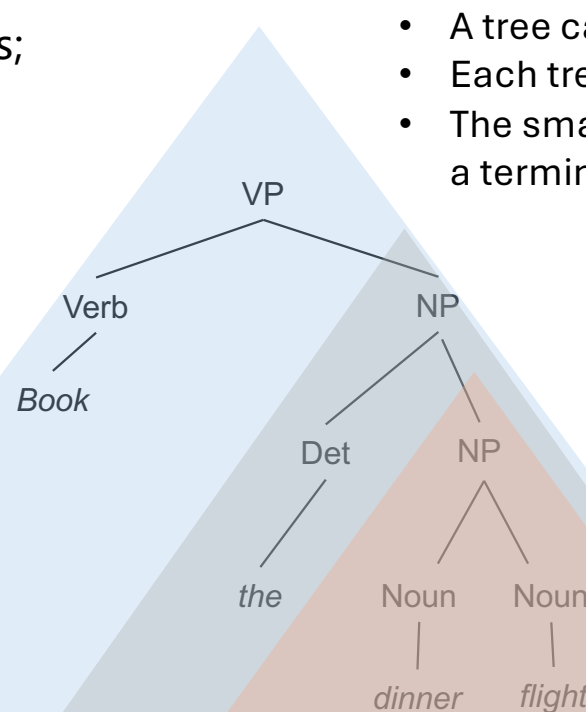
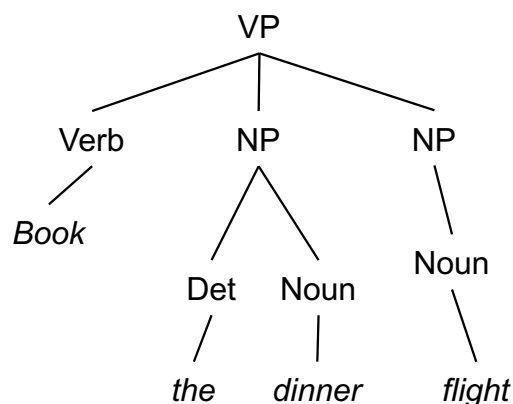
Probabilistic CFG

- Motivations

- CYK algorithm only finds all parsing trees of a given sentence, but can't

- evaluate the probability of a parsing tree;
 - tell which tree is more likely than another trees;
 - identify the optimal tree.

- Example: parse "*book the dinner flight*"



- A tree can contain sub-trees.
- Each tree is a constitute.
- The smallest trees have a terminal only.

Probabilistic CFG

- Assign probability to each production in a CFG G
 - The probabilities of the rules with the same left-hand-side sum to one.

sum to 1 $\leftarrow$

$S \rightarrow NP VP$	1.0	$NP \rightarrow NP PP$	0.4
$PP \rightarrow P NP$	1.0	$NP \rightarrow \textit{astronomers}$	0.1
$VP \rightarrow V NP$	0.7	$NP \rightarrow \textit{ears}$	0.18
$VP \rightarrow VP PP$	0.3	$NP \rightarrow \textit{saw}$	0.04
$P \rightarrow \textit{with}$	1.0	$NP \rightarrow \textit{stars}$	0.18
$V \rightarrow \textit{saw}$	1.0	$NP \rightarrow \textit{telescopes}$	0.1

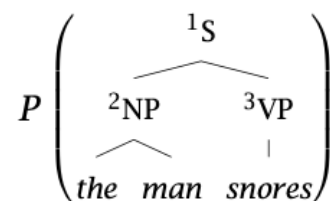
N^j is a non-terminal and can be the start symbol S .

- Want to find the probability of any derivation $N^j \xRightarrow{*} w_a \dots w_b$

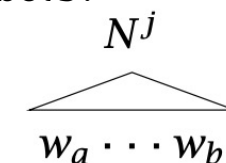
$$\Pr(N^j \xRightarrow{*} w_a \dots w_b | G)$$

- In particular, prob of a sentence is

$$\Pr(S \xRightarrow{*} w_a \dots w_b | G)$$



$$= P({}^1S_{13} \rightarrow {}^2NP_{12} {}^3VP_{33}, {}^2NP_{12} \rightarrow \textit{the}_1 \textit{man}_2, {}^3VP_{33} \rightarrow \textit{snores}_3)$$



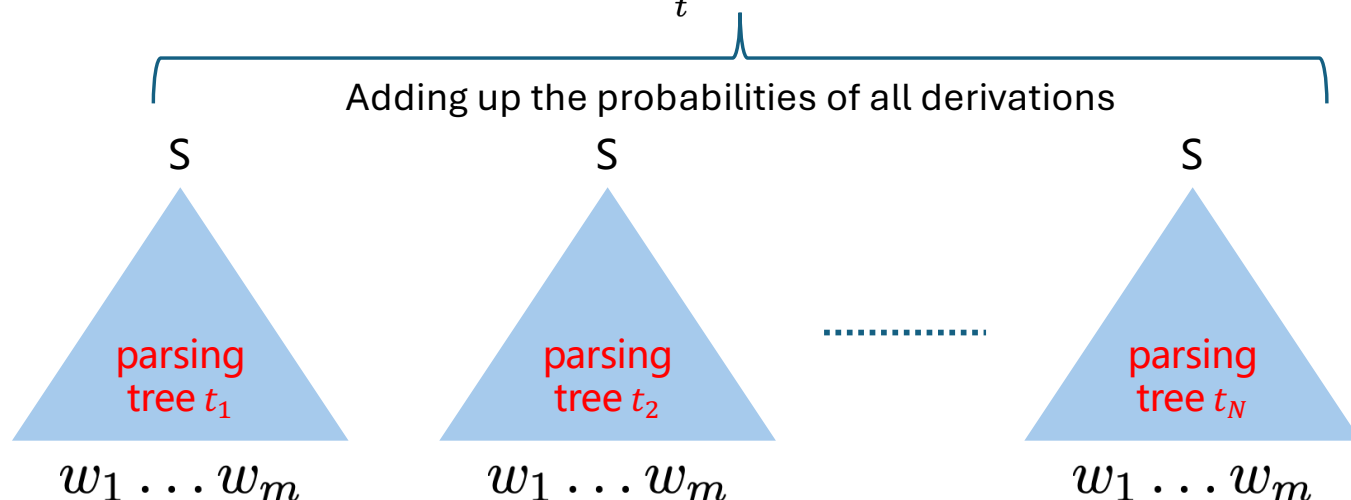
$w_a \dots w_b$ are some terminals

Probabilistic CFG

- The probability of a parse tree t for the sentence $w_1 \dots w_m$ given the grammar G is $\Pr(t, w_1 \dots w_m | G)$
- The probability of the sentence $w_1 \dots w_m$ is

$$\Pr(w_1 \dots w_m | G) = \sum_t \Pr(t, w_1 \dots w_m | G)$$

marginalization out t
by total probability



There are N finitely many number of trees that can generate $w_1 \dots w_m$

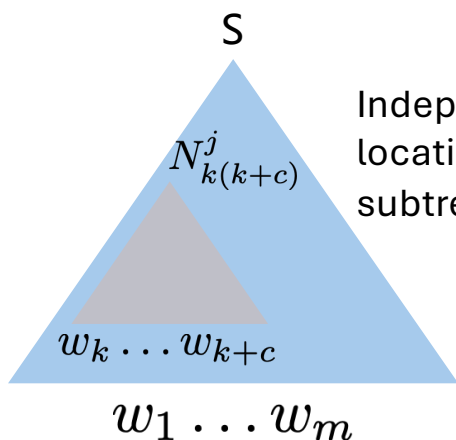
But N can be exponentially large.

Probabilistic CFG

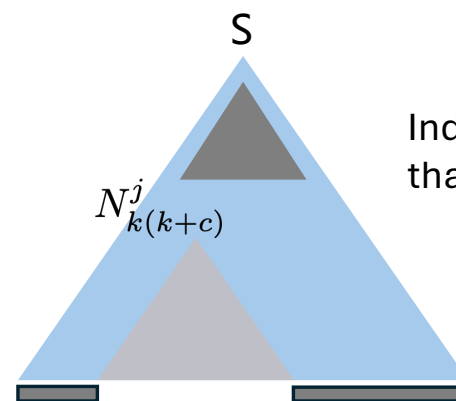
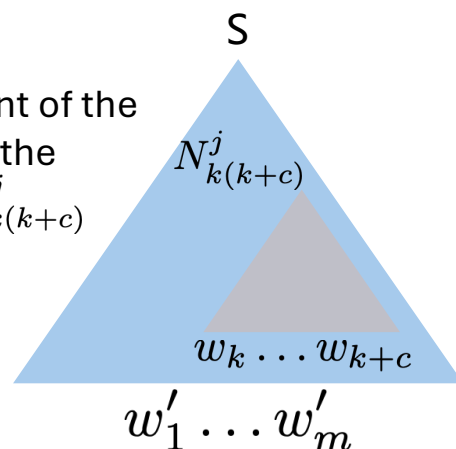
Notation: $N_{k(k+c)}^j$ means the non-terminal j derives words starting from k and ending at $k + c$

- Need to make some assumption about the CFG to develop an approximation of the generation probabilities, leading to more tractable algorithms.

1. Place invariance $\forall k \ P(N_{k(k+c)}^j \rightarrow \zeta)$ is the same
2. Context-free $P(N_{kl}^j \rightarrow \zeta | \text{anything outside } k \text{ through } l) = P(N_{kl}^j \rightarrow \zeta)$
3. Ancestor-free $P(N_{kl}^j \rightarrow \zeta | \text{any ancestor nodes outside } N_{kl}^j) = P(N_{kl}^j \rightarrow \zeta)$



Independent of the location of the subtree $N_{k(k+c)}^j$



Independent of the tree that generate $N_{k(k+c)}^j$

Independent of context outside the tree at $N_{k(k+c)}^j$

Probabilistic CFG

- Example to demonstrate place invariant and context- and ancestor-free PCFG

$$P \left(\begin{array}{cc} & {}^1S \\ & \diagdown \quad \diagup \\ {}^2NP & {}^3VP \\ \diagdown \quad \diagup & | \\ the \ man & snores \end{array} \right)$$

$^1S_{13}$: the index 1 in the left superscript just to label some non-terminal in case there are multiple non-terminals that have the same names in the derivation.

$$\begin{aligned}
&= P({}^1S_{13} \rightarrow {}^2NP_{12} {}^3VP_{33}, {}^2NP_{12} \rightarrow the_1 man_2, {}^3VP_{33} \rightarrow snores_3) \\
&= P({}^1S_{13} \rightarrow {}^2NP_{12} {}^3VP_{33})P({}^2NP_{12} \rightarrow the_1 man_2 | {}^1S_{13} \rightarrow {}^2NP_{12} {}^3VP_{33}) \\
&\quad P({}^3VP_{33} \rightarrow snores_3 | {}^1S_{13} \rightarrow {}^2NP_{12} {}^3VP_{33}, {}^2NP_{12} \rightarrow the_1 man_2) \\
&= P({}^1S_{13} \rightarrow {}^2NP_{12} {}^3VP_{33})P({}^2NP_{12} \rightarrow the_1 man_2)P({}^3VP_{33} \rightarrow snores_3) \\
&= P(S \rightarrow NP VP)P(NP \rightarrow the man)P(VP \rightarrow snores)
\end{aligned}$$

(Conditional probability)

(Don't care about ancestors
and context in the conditions)

(Place invariance to remove the indices for locations)

| Inference and estimation algorithms with PCFG

- Find the probability of a sentence

$$\Pr(w_1 \dots w_m | G) = \sum_t \Pr(t, w_1 \dots w_m | G)$$

- Find the most likely parsing tree

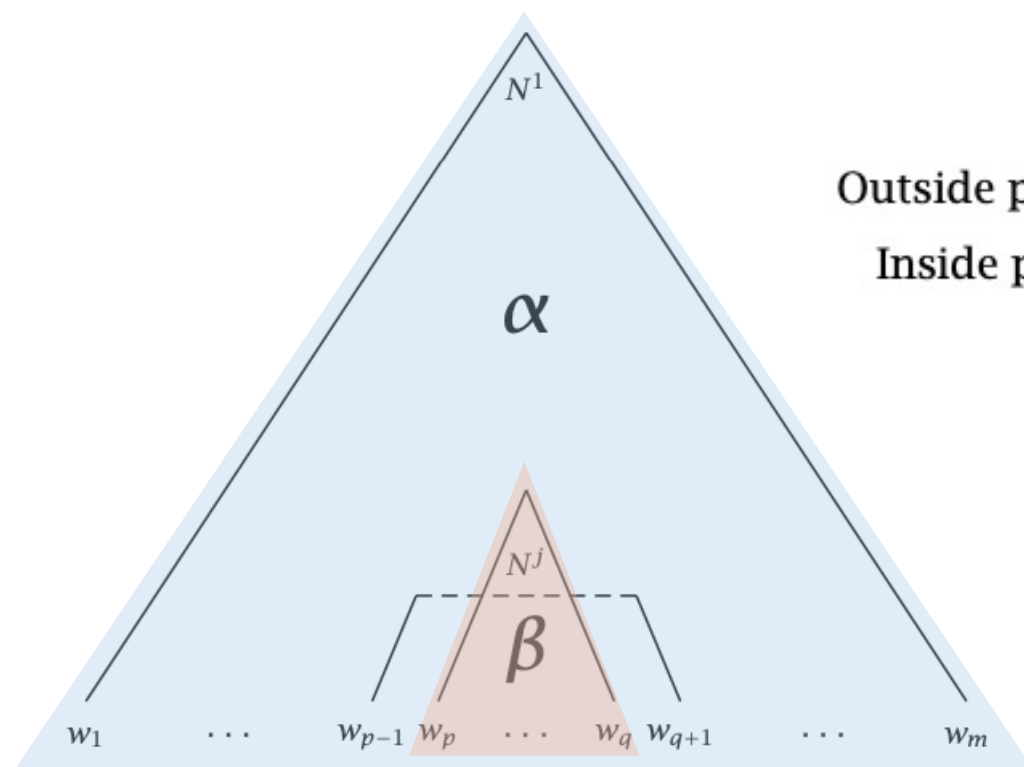
$$t^* = \arg \max_t \Pr(t | w_1 \dots w_m, G)$$

- Learn a PCFG from a training corpus using MLE.

$$G^* = \arg \max_G \Pr(w_1 \dots w_m | G)$$

Inside and outside probabilities

- Similar to HMM's forward and backward probabilities



Outside probability $\alpha_j(p, q) = P(w_{1(p-1)}, N_{pq}^j, w_{(q+1)m} | G)$

Inside probability $\beta_j(p, q) = P(w_{pq} | N_{pq}^j, G)$

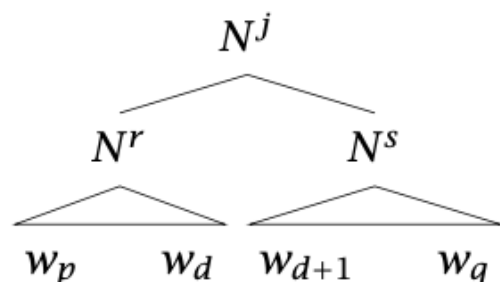
Notation: N_{pq}^j means the non-terminal j derives words starting from p and ending at q

Inside algorithm using DP

- Sentence probability derived from a PCFG $P(w_{1m}|G) = P(N^1 \xRightarrow{*} w_{1m}|G)$
- Base case $= P(w_{1m}|N^1_{1m}, G) = \beta_1(1, m)$

$$\begin{aligned}\beta_j(k, k) &= P(w_k|N^j_{kk}, G) \\ &= P(N^j \rightarrow w_k|G)\end{aligned}$$

- Induction due to CNF
 - CYK with probabilities



$$\begin{aligned}\beta_j(p, q) &= P(w_{pq}|N^j_{pq}, G) \\ &= \sum_{r,s} \sum_{d=p}^{q-1} P(w_{pd}, N^r_{pd}, w_{(d+1)q}, N^s_{(d+1)q}|N^j_{pq}, G) \quad \text{non-terminals for two children} \\ &= \sum_{r,s} \sum_{d=p}^{q-1} P(N^r_{pd}, N^s_{(d+1)q}|N^j_{pq}, G) P(w_{pd}|N^r_{pd}, N^s_{(d+1)q}, G) \\ &\quad \times P(w_{(d+1)q}|N^j_{pq}, N^r_{pd}, N^s_{(d+1)q}, w_{pd}, G) \quad \text{all possible breaking point d} \\ &= \sum_{r,s} \sum_{d=p}^{q-1} P(N^r_{pd}, N^s_{(d+1)q}|N^j_{pq}, G) P(w_{pd}|N^r_{pd}, G) \\ &\quad \times P(w_{(d+1)q}|N^s_{(d+1)q}, G) \quad \text{ancestor- and context-free} \\ &= \sum_{r,s} \sum_{d=p}^{q-1} P(N^j \rightarrow N^r N^s) \beta_r(p, d) \beta_s(d+1, q) \quad \text{place invariant}\end{aligned}$$

Inside probabilities: example

$S \rightarrow NP VP$	1.0	$NP \rightarrow NP PP$	0.4
$PP \rightarrow P NP$	1.0	$NP \rightarrow \textit{astronomers}$	0.1
$VP \rightarrow V NP$	0.7	$NP \rightarrow \textit{ears}$	0.18
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$V \rightarrow \textit{saw}$	1.0	$NP \rightarrow \textit{telescopes}$	0.1

	1	2	3	4	5
1	$\beta_{NP} = 0.1$		$\beta_S = 0.0126$		$\beta_S = 0.0015876$
2		$\beta_{NP} = 0.04$ $\beta_V = 1.0$	$\beta_{VP} = 0.126$		$\beta_{VP} = 0.015876$
3			$\beta_{NP} = 0.18$		$\beta_{NP} = 0.01296$
4				$\beta_P = 1.0$	$\beta_{PP} = 0.18$
5					$\beta_{NP} = 0.18$
	<i>astronomers</i>	<i>saw</i>	<i>stars</i>	<i>with</i>	<i>ears</i>

Span (2,5) "saw stars with ears":

Span (2,3) "saw stars":

$$VP \rightarrow V NP \Rightarrow 0.7 \times 1.0 \times 0.18 = \mathbf{0.126}$$

Span (4,5) "with ears":

$$PP \rightarrow P NP \Rightarrow 1.0 \times 1.0 \times 0.18 = \mathbf{0.18}$$

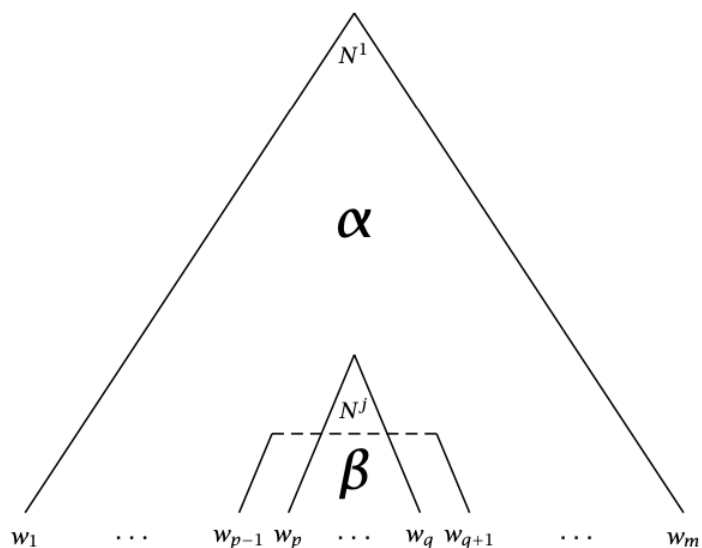
$$VP \rightarrow V(2,2) NP(3,5) \Rightarrow 0.7 \times 1.0 \times 0.01296 = 0.009072$$

$$VP \rightarrow VP(2,3) PP(4,5) \Rightarrow 0.3 \times 0.126 \times 0.18 = 0.006804$$

$$\beta_{VP}(2,5) = 0.009072 + 0.006804 = \mathbf{0.015876}$$

Outside probability

- Alternative to inside probability, the probability of a sentence given a PCFG can be calculated using outside probability.



$$\begin{aligned}
 P(w_{1m}|G) &= \sum_j P(w_{1(k-1)}, w_k, w_{(k+1)m}, N_{kk}^j | G) \\
 &= \sum_j \boxed{P(w_{1(k-1)}, N_{kk}^j, w_{(k+1)m} | G)} \leftarrow \text{outside prob.} \\
 &\quad \times P(w_k | w_{1(k-1)}, N_{kk}^j, w_{(k+1)m}, G) \\
 &= \sum_j \alpha_j(k, k) P(N^j \rightarrow w_k) \quad \text{context-free and place invariant}
 \end{aligned}$$

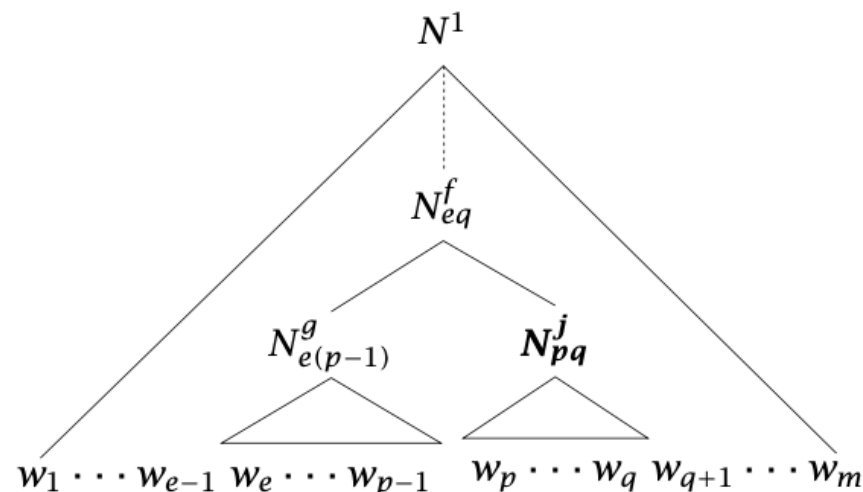
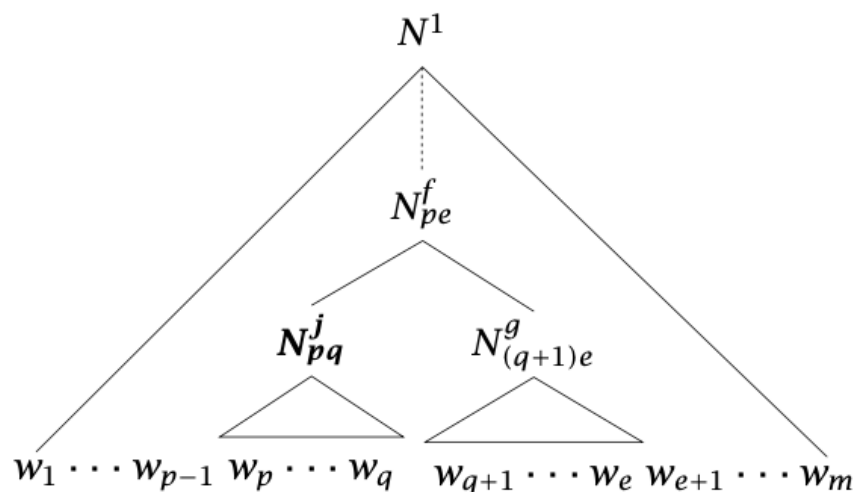
More generally, define $\alpha_j(p, q)$ as the outside probability of deriving a non-terminal N^j at some point and the words outside the range of (p, q) .

Inside algorithm using DP

- Compute the outside probabilities using dynamic programming.
- Base case

$\alpha_1(1, m) = 1$	Nothing outside for sure, so the probability of starting from non-terminal 1 (designated as the starting symbol) is 1.
$\alpha_j(1, m) = 0 \text{ for } j \neq 1$	
- Induction due to CNF: the target outside probability can be for

○ N_{pq}^j is the root of the left child	N_{pq}^j is the root of the right child
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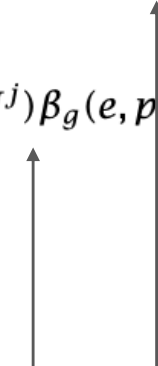


Inside algorithm using DP

- Dynamic programming: subproblems for the parent and the sibling are solved. The sub-problems are created by conditioning on:
- the **left** or **right** child, and the **end** or **starting** point of the sibling.

$$\begin{aligned}
 \alpha_j(p, q) &= \left[\sum_{f, g \neq j} \sum_{e=q+1}^m P(w_{1(p-1)}, w_{(q+1)m}, N_{pe}^f, N_{pq}^j, N_{(q+1)e}^g) \right] \\
 &\quad + \left[\sum_{f, g} \sum_{e=1}^{p-1} P(w_{1(p-1)}, w_{(q+1)m}, N_{eq}^f, N_{e(p-1)}^g, N_{pq}^j) \right] \\
 &= \left[\sum_{f, g \neq j} \sum_{e=q+1}^m P(w_{1(p-1)}, w_{(e+1)m}, N_{pe}^f) P(N_{pq}^j, N_{(q+1)e}^g | N_{pe}^f) \right. \\
 &\quad \times P(w_{(q+1)e} | N_{(q+1)e}^g) \left. \right] + \left[\sum_{f, g} \sum_{e=1}^{p-1} P(w_{1(e-1)}, w_{(q+1)m}, N_{eq}^f) \right. \\
 &\quad \times P(N_{e(p-1)}^g, N_{pq}^j | N_{eq}^f) P(w_{e(p-1)} | N_{e(p-1)}^g) \left. \right]
 \end{aligned}$$

$$\begin{aligned}
 &= \left[\sum_{f, g \neq j} \sum_{e=q+1}^m \alpha_f(p, e) P(N^f \rightarrow N^j N^g) \beta_g(q+1, e) \right] \\
 &\quad + \left[\sum_{f, g} \sum_{e=1}^{p-1} \alpha_f(e, q) P(N^f \rightarrow N^g N^j) \beta_g(e, p-1) \right]
 \end{aligned}$$



 Need inside probabilities

Probability of a sentence

- Suppose the inside and outside probabilities are computed at some non-leaf node that is the non-terminal j and spans $[p, q]$.
- Then the probability of the sentence with that leaf node is:

$$\begin{aligned}\alpha_j(p, q)\beta_j(p, q) &= P(w_{1(p-1)}, N_{pq}^j, w_{(q+1)m} | G) P(w_{pq} | N_{pq}^j, G) \\ &= P(w_{1m}, N_{pq}^j | G)\end{aligned}$$

- The probability of the sentence with some non-terminal spanning $[p, q]$:

$$P(w_{1m}, N_{pq} | G) = \sum_j \alpha_j(p, q)\beta_j(p, q)$$

- but this is just the probability of the sentence when there is non-terminal j deriving words p to q .

- Two special cases: $P(w_{1m} | G) = P(N^1 \xRightarrow{*} w_{1m} | G)$ or $= \sum_j \alpha_j(k, k) P(N^j \rightarrow w_k)$
 $= P(w_{1m} | N_{1m}^1, G) = \beta_1(1, m)$

Find the optimal parsing tree

- Finding the optimal tree is called “decoding”, similar to the Viterbi algorithm for HMM.
- Viterbi $\approx$ forward, CYK $\approx$ optimal-tree-finding
- Filling out the CYK matrix:
 - On the diagonal $\delta_i(p, p) = P(N^i \rightarrow w_p)$
 - In cell (p, q) spanning more than 1 words:
 - max probability $\psi_i(p, q) = \arg \max_{(j,k,r)} P(N^i \rightarrow N^j N^k) \delta_j(p, r) \delta_k(r + 1, q)$
 - record the backpointers $\delta_i(p, q) = \max_{\substack{1 \leq j, k \leq n \\ p \leq r < q}} P(N^i \rightarrow N^j N^k) \delta_j(p, r) \delta_k(r + 1, q)$
 - At the top-right corner $P(\hat{t}) = \delta_1(1, m)$
- Reconstructing the optimal tree by back-tracing using the back-pointers.
 - if N_{pq}^i is in the Viterbi parse and $\psi_i(p, q) = (j, k, r)$, then left child = N_{pr}^j and right child = $N_{(r+1)q}^k$

Running example

It is similar to the Inside Algorithm, replacing the sum with max

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	<i>astronomers</i>	<i>saw</i>	<i>stars</i>	<i>with</i>	<i>ears</i>

Span (2,3) "saw stars":

$$VP \rightarrow V NP \Rightarrow 0.7 \times 1.0 \times 0.18 = \mathbf{0.126}$$

Span (4,5) "with ears":

$$PP \rightarrow P NP \Rightarrow 1.0 \times 1.0 \times 0.18 = \mathbf{0.18}$$

These spans have no alternatives to maximize.

Span (2,5) "saw stars with ears":

$$VP \rightarrow V(2,2) NP(3,5) \Rightarrow 0.7 \times 1.0 \times 0.01296 = 0.009072$$

$$VP \rightarrow VP(2,3) PP(4,5) \Rightarrow 0.3 \times 0.126 \times 0.18 = 0.006804$$

$$\delta_{VP}(2,5) = \max(0.009072, 0.006804) = 0.009072$$

$\psi_{VP}(2,5) = (V, NP, 2)$: parsing span (2,5) with root VP,
left child rooted at V, right child rooted at NP
and breaking the span (2,5) at the position 2.

| Extra readings

- For PCFG, see Chap 11 of FSNLP.
- For parsing with PCFG, see Chap 12 of FSNLP.

| Demon

- Use a toy PCFG to demonstrate the Inside and Outside algorithm.

| Conclusion

- PCFG is a probabilistic version of CFG, allowing one to find optimal parsing trees out of many possible trees.
- Inside and outside algorithm for calculating probability of a sentence.
- CYK-like algorithm for finding the optimal parsing tree.

Quiz

1. Pen and paper; no compute or cellphone allowed.
 2. Turn in your answer sheet when you leave the classroom.
- Q1 (T/F): In a PCFG, the probabilities of the rules/products with the same left-hand-side should sum to 1.
 - Q2 (T/F): Inside algorithm is a dynamic programming algorithm.
 - Q3 (T/F): When running the outside algorithm, one does not need the results of the inside algorithm on the same input sentence.
 - Q4 (T/F): To find the optimal parsing tree using the CYK algorithm, the PCFG must be in CNF.
 - A: T, T, F, T